

Ask Health Pro

AI-Enabled Intelligent Triage System for Enhanced Healthcare Accessibility

1. ABOUT US

GAUDE BUSINESS AND INFRASTRUCTURE SOLUTIONS PVT LTD is a strategic business unit committed to delivering proactive value to clients through tailored, high-quality solutions. Operating from Technopark Campus, Trivandrum, under Kerala Start-Up Mission, GAUDE provides software development and managed services supported by a globally experienced management and technical team. The organization focuses on Application Development, Maintenance, and Managed Services, consistently ensuring optimized delivery, innovation, and quality that drive measurable impact for its clients.

2. EXECUTIVE SUMMARY

The "Ask Health Pro" initiative presents a transformative opportunity to revolutionize initial patient interaction and triage within healthcare systems. The key challenge lies in effectively managing and prioritizing incoming patient inquiries, ensuring timely access to appropriate medical professionals. GAUDE proposes an AI-driven intelligent triage system that streamlines patient intake, accurately classifies cases, and facilitates seamless communication between patients and doctors. This solution is expected to significantly enhance operational efficiency, improve patient satisfaction, and optimize resource allocation across the healthcare organization. Through collaborative partnership, GAUDE will deliver a cutting-edge solution that reshapes the landscape of patient care and accessibility.

3. PROBLEM STATEMENT

The current patient intake and triage process often presents significant operational bottlenecks for healthcare providers. The reliance on manual methods for symptom collection, case classification, and doctor assignment leads to inefficiencies, delays, and inconsistent patient experiences. The absence of a centralized system for managing patient information and medical data further complicates the process, requiring administrative staff to spend considerable time on repetitive tasks. This disjointed approach strains resources, increases wait times for patients, and can potentially impact the quality and timeliness of medical care.

The lack of an efficient and scalable solution for patient triage and lead generation poses a strategic risk for healthcare organizations. Delayed or inaccurate initial assessments can result in misallocation of resources, longer patient wait times, and potential dissatisfaction among patients. In a competitive healthcare landscape, the inability to effectively manage

patient inquiries and convert them into qualified leads hinders growth and operational sustainability. Addressing these challenges is critical to ensure optimal resource utilization, enhanced patient satisfaction, and a proactive approach to healthcare delivery.

4. PROPOSED SOLUTION

GAUDE proposes an AI-driven intelligent triage system, "Ask Health Pro," leveraging advanced technologies to streamline the patient intake process and optimize resource allocation. This solution integrates cutting-edge AI capabilities with robust communication and data management tools to create a cohesive and efficient healthcare ecosystem. The technology landscape encompasses AI-powered symptom analysis, intelligent case classification, secure data storage, and seamless communication channels, all designed to enhance patient access and improve doctor efficiency.

GAUDE's internal solutions will be integrated to enhance the Ask Health Pro project. This includes voice-to-text capabilities from Convo AI for verbal symptom descriptions, sentiment analysis to prioritize urgent cases, multi-agent routing from Helpybo Multi Agents Builder for dynamic case allocation, secure document processing and access control via Hyperledger, and Convo AI's reporting and analytics dashboard for performance tracking. In addition, AI-Powered MCQ Generator ensures comprehensive and efficient information gathering. These features will collectively enhance accessibility, streamline triage, and ensure patient privacy and security.

"Ask Health Pro" offers a strategic advantage by automating the initial triage process, enabling healthcare providers to focus on delivering expert medical advice and treatment. The solution's differentiation lies in its AI-driven capabilities, seamless communication channels, and comprehensive data management tools, all tailored to meet the unique needs of the healthcare industry. By aligning with the client's objectives of improved patient engagement, streamlined administrative tasks, and enhanced overall efficiency, "Ask Health Pro" empowers healthcare organizations to deliver superior patient care and optimize resource utilization.

5. KEY FEATURES AND FUNCTIONALITIES

The proposed solution encompasses key functionalities tailored to address the client's operational and technical needs.

AI-Powered Symptom Assessment: This feature utilizes advanced AI algorithms to engage users in conversational interactions, accurately collecting detailed information about their symptoms and medical history. This capability streamlines the triage process by automating initial patient assessments.

Intelligent Case Classification: This feature leverages machine learning models to intelligently classify user cases into appropriate medical departments based on the

collected symptoms and medical information. This facilitates efficient case allocation to the most suitable doctors.

Automated Case Sheet Generation: The system automatically generates comprehensive case sheets that summarize user information, symptoms, and medical history, providing doctors with a clear and concise overview of each case. React Hook Form ensures data integrity and a seamless user experience.

Efficient Case Allocation to Doctors: This feature enables administrators to efficiently allocate cases to doctors based on their specialization, availability, and user location, optimizing resource utilization and reducing patient wait times. Multi-agent routing from Helpybo Multi Agents Builder further enhances this process.

Secure Doctor-User Communication Channel: The system facilitates secure and reliable communication between doctors and users through integrated WhatsApp/email channels, ensuring patient privacy and confidentiality. Supabase Realtime allows for live doctor-user communication within the case sheet.

Secure Medical File Storage: Leveraging Amazon S3, the system provides secure and scalable storage for uploaded medical files, such as X-rays, with appropriate access control policies to ensure HIPAA compliance. Hyperledger's document processing feature adds further security with audit trails.

Sentiment Analysis and Prioritization: Integrated sentiment analysis from Convo AI analyzes user responses to identify and prioritize urgent cases, ensuring timely attention for patients experiencing high levels of distress or anxiety.

Semantic Search on Medical Records: Utilizing Pinecone, a vector database, the system enables semantic search across stored medical records, allowing doctors to quickly find relevant cases or medical literature based on meaning. This enhances the efficiency of diagnosis and treatment planning.

6. IMPLEMENTATION APPROACH AND DEVELOPMENT PROCESS

The implementation will follow a phased approach to ensure scalability and seamless adoption.

Phase I – Foundation and Core Development: This phase focuses on establishing the core infrastructure and functionalities of the "Ask Health Pro" system. Key activities include developing the AI-powered symptom assessment module, implementing the intelligent case classification algorithm, and setting up secure medical file storage using Amazon S3. The timeframe for this phase is estimated at 2-3 months.

Phase II – Integration and Enhancement: This phase involves integrating GAUDE's internal solutions, such as Convo AI and Helpybo Multi Agents Builder, to enhance the system's capabilities. Additionally, external integrations with communication channels like WhatsApp/email and data management tools like Google Cloud Healthcare API will be implemented. The timeframe for this phase is estimated at 3-4 months.

Phase III – Optimization and Expansion: This phase focuses on optimizing the system's performance, enhancing user experience, and expanding its functionalities based on user feedback and evolving requirements. Key activities include refining the AI algorithms, implementing real-time communication features, and integrating analytics dashboards for performance tracking. The timeframe for this phase is estimated at 2-3 months.

This phased execution ensures quality, adaptability, and stakeholder confidence.

7. EXPECTED OUTCOMES AND BENEFITS

The implementation of "Ask Health Pro" is expected to deliver significant operational outcomes for the client. The AI-driven triage process will streamline patient intake, reduce administrative burden, and optimize resource allocation. The intelligent case classification and automated case sheet generation will improve the accuracy and efficiency of initial patient assessments, leading to better healthcare outcomes. The system's scalability will enable the client to handle increasing patient volume without compromising service quality.

The implementation of "Ask Health Pro" will provide multifaceted benefits for key stakeholder groups. End users will experience improved access to medical professionals through a streamlined triage process. Administrators will benefit from reduced administrative overhead, automated case allocation, and improved communication facilitation. Management will gain access to valuable insights through reporting and analytics dashboards, enabling data-driven decision-making and continuous improvement of healthcare delivery.

8. CONCLUSION

"Ask Health Pro" represents a transformative solution that will redefine the patient intake process and optimize resource allocation for healthcare organizations. The AI-driven intelligent triage system will streamline operations, improve patient satisfaction, and enhance the overall efficiency of healthcare delivery. GAUDE is committed to a partnership-driven approach, leveraging its expertise and innovative solutions to deliver a cutting-edge system tailored to the client's specific needs. We are ready to embark on the next steps, collaborating closely with the client to bring this vision to reality and revolutionize healthcare accessibility.